

The Cost of Stability: Evaluation of 6T, 8T and 10T SRAM Cells

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Abstract

With the trends of device scaling and increased demand for cache, SRAM cell stability has become a critical problem. Specifically, cell stability requires the SRAM to be functional and stable during both write and read events. In the traditional 6T SRAM architecture, there is a trade off between the stability of read and write events. Cell stability during read requires strong pull down and weak access transistors. On the other hand, successful write requires weak pull up and strong access transistors. This issue is further impacted by the decrease in SNM brought by the scaling of SRAMs. In this poster, we evaluate and compare the 6T SRAM with higher transistor designs of SRAM as well as the impact of technology scaling.

6T SRAM

The 6T SRAM cell consists of two cross coupled inverters and two passgates. Cell stability during read events requires strong pull down and weak passgates (high β ratio). On the other hand, successful write events requires weak pull up and strong passgates (low γ ratio). The constraint on the pass gate leads to the fundamental tradeoff.

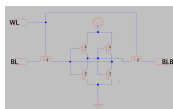


Figure 1

8T SRAM

The 8T SRAM cell shown in fig 2. The 8T SRAM cell maintains the core topology of the 6T SRAM cell but adds a read access transistor controlled by RWL in series with a pull-down transistor whose gate is driven by the storage node. This track connects to a different RBL. This configuration eliminates the cell disturbs during read access at the cost of added complexity and area.¹

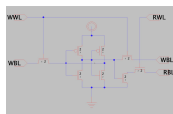


Figure 2

10T SRAM

The 10T cell extends the 8T idea by adding two PMOS gating devices to the read path, giving the storage node stronger isolation during a read. This provides a small additional read-margin improvement and, more importantly, allows reliable operation at lower voltages² where the 8T cell begins to fail³.

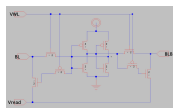
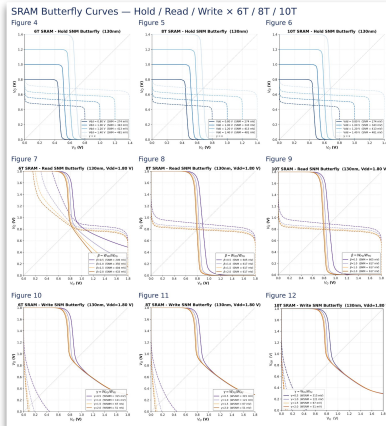


Figure 3

Methodology

The cell schematics were produced using the Electric VLSI Design System and exported as a SPICE file. Simulations were performed using NgSpice, with the results written to .data files. The results are plotted in python. The SkyWater 130nm PDK was used in both the Cell Stability Evaluation section and the Technology Scaling section. In the Technology Scaling section, the 22nm PDK used the PTM that came from ASU. The 32nm, 45nm, 65nm, 90nm and 180nm PTM PDKs were obtained from the Microelectronics Co-design Research Group from the University of Minnesota.

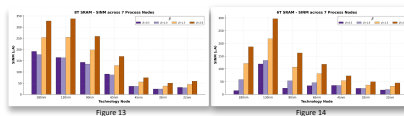
Cell Stability Evaluation



Write stability is primarily independent of design because the write mechanisms are the same (write butterfly curves in Figs. 10–12). The read butterfly underlines the advantage of 8T and 10T over the 6T SRAM (read butterfly curves in Figs. 7–9) due to its decoupled read mechanism. The hold SNM curves are constant for a given voltage across the different SRAM designs. This is because the hold SNM concerns with the storage node of the cross coupled inverter, which is the same across the designs.

NCurves

Fig 13 and 14 illustrate the SINM changes for a 6T and 8T SRAM cell. A similar trend becomes apparent: As technology node scales down the SINM generally decreases. On the contrary, increasing beta consistently improves SINM with $\beta = 2.0$ giving the highest value. Since the 8T cell has a separate read path from the storage cell, it maintains a stronger SINM.



Technology Scaling

Across all technology nodes, with the exception of 22nm, hold SNM scales with Vdd. The reason that this trend doesn't hold for 22nm is that the gate oxide likely cannot sustain the high voltage beyond 0.8V. When nominal Vdd for each process node is applied, the hold SNM will be larger for higher nm nodes due to the effect of Vdd on Hold SNM. The results are the same for 8T and 10T.

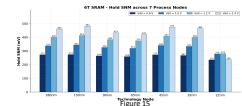


Figure 15

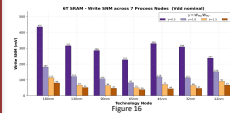


Figure 16

For the read SNM, we see that the trend of increasing read margin with increasing beta ratio. The read SNM generally goes down as the technology node decreases. It is important to note the different PDKs used in this table. Within PDKs of the same source, we see the decrease in read SNM is evident in the 180nm, 90nm and 65nm is evident as well as in the 45nm, 32nm and 22nm nodes.

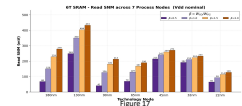


Figure 17

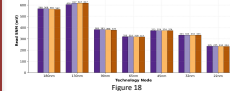


Figure 18

In the write SNM, we apply the appropriate nominal Vdd for each technology node. The relation of low gamma and high write static noise margin holds across all technology nodes. The write SNM values of the 6T SRAM cell are the same for 8T and 10T across technology nodes given the same gamma. However, it is important to note that the separation of the write and read lines in 8T and 10T allows for a high gamma ratio without compromising the read SNM. There is a general downward write SNM trend as technology node scales down. The reason for the jump at 45nm is because starting at 45nm PTM separates into low power and high performance device flavors.

In contrast to the previous two comparisons, the Read SNM for the 8T and 10T are not the same as that of the 6T. This is because the read mechanism in the 8T SRAM is decoupled. This leads to a read SNM comparable to the hold SNM.

Tradeoff and Key Takeaways

The analysis confirms the main trends with regards to beta and gamma relation to the static noise margins with read and write for 6T, 8T and 10T cells. The 8T and 10T cells show superior read margins due to its read decoupling across all technology scalings. On the other hand, the hold and write SNMs are the same for 6T, 8T and 10T across scaling. The main takeaways is that our data confirms that the read and write SNM generally decreases as technology node decreases, motivating for optimization methods to increase SNM. The fundamental tradeoff for 6T between the read and write margin motivates higher transistor count designs including 8T and 10T. Our analysis show that the decoupling of the read lines allows the write event to be optimized.

Sources

- [1] L. Chang et al., "An 8T-SRAM for Variability Tolerance and Low-Voltage Operation in High-Performance Caches," IEEE Journal of Solid-State Circuits, vol. 43, no. 4, pp. 956–963, Apr. 2008, doi: <https://doi.org/10.1109/JSSC.2007.917509>.
- [2] H. Pahuja, M. Tyagi, S. Pandey, and B. Singh, "A 16 nm Robust DG-FinFET Based 10T Static Random-Access Memory Cell Design for Ultra-Low Power Memory Design," Advanced Science, Engineering and Medicine, vol. 9, no. 12, pp. 1022–1028, Dec. 2017, doi: <https://doi.org/10.1155/assem.2017.2082>.
- [3] M. Sharma, S. S. Gill, and K. Sharma, "Comparative Analysis 6T, 8T and 10T SRAM in CMOS and FinFET Technologies: A Review," 2024 International Conference on Computer, Electronics, Electrical Engineering & their Applications (ICCEE3), pp. 1–6, Jun. 2024, doi: <https://doi.org/10.1109/ic2e362166.2024.10827672>.